

Multiple Choice (10 marks)

1. On a Cartesian coordinate plane, points $(1, 2)$ and $(7, 4)$ are opposite vertices of a square. What is the area of the square?
 - (a) 20
 - (b) 34
 - (c) 40
 - (d) $\sqrt{40}$
2. If we take $\sin x$ and change it to $3 \sin 2x$, what will happen to the domain and range of the function?
 - (a) Domain and range remain the same
 - (b) Domain is double; range remains the same
 - (c) Domain is tripled; range is doubled
 - (d) Domain is the same; range is tripled
3. Write 8 and 2 over 4 as an improper fraction in simplest form.
 - (a) 17 over 2
 - (b) 34 over 4
 - (c) 17 over 4
 - (d) 19 over 2
4. Lucy is counting by 2's. She starts with the number 2 and stops at the number 50. Which number would Lucy not count?
 - (a) 11
 - (b) 22
 - (c) 34
 - (d) 48
5. If 50% of a number is 20, what is 75% of the number?
 - (a) 8
 - (b) 15
 - (c) 30
 - (d) 45

True / False (10 marks)

Answer True or False

6. True or False: Adding the number 6 to the list will increase the mean of the numbers.
7. True or False: The least common multiple of 4 and 10 is 40.
8. True or False: The number 9 multiplied by 4 equals 36.
9. True or False: Darius has 18 cookies, which is three times the number of cookies Sara has.
10. True or False: The total number of pieces in 3 puzzles, each with 1,250 pieces, is 4,650 pieces.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Four concentric circles are drawn with radii of 1, 3, 5 and 7. The inner circle is painted black, the ring around it is white, the next ring is black and the outer ring is white. What is the ratio of the black area to the white area? Express your answer as a common fraction.
12. Discuss the method to determine the greatest common divisor (GCD) of the numbers 54, 36, and 24. Explain your reasoning and any steps you take to reach your conclusion.

End of Paper